Journal of Experimental Social Psychology 128 (2027) 105015



Contents lists available at ScienceDirect

Journal of Experimental Social Psychology

journal homepage: www.elsevier.com/locate/jesp



Bimodal emotion decoding: When faces and voices diverge

Hweemin Tan [a], Kerry Kawakami [b], Jinpeng Fu [a], Shuoan Li [a], Zhihe Pan [a], Xia Fang [a,c,*]

[a] Department of Psychology and Behavioral Sciences, Zhejiang University, PR China
[b] Department of Psychology, York University, Canada
[c] Zhejiang Key Laboratory of Neurocognitive Development and Mental Health, Zhejiang University, PR China

ARTICLE INFO

Editor name: Dr. Rachael Jack

Keywords:
Bimodal emotion decoding
Incongruent emotional cues
Facial dominance
Vocal dominance
Mixed emotions

ABSTRACT

It is well established that people infer emotions from both faces and voices, yet how they integrate these cues when the two modalities convey conflicting emotions remains unclear. Evidence for facial dominance has typically come from studies contrasting facial and vocal expressions of anger and happiness, often using binary response formats that require observers to choose between face-consistent andvoice-consistent emotion labels. These findings, however, fail to inform us whether facial dominance reflects a general principle of bimodal emotion perception or whether these results are shaped by the specific emotion pairing and response format. Across six preregistered experiments (total $N = 359$), participants judged dynamic face–voice expressions spanning three incongruent pairings: anger–happiness, anger–sadness, and happiness–sadness. Experiments 1–3 used multidimensional ratings, and Experiments 4–6 used open-ended descriptions. The results revealed a pairing-dependent pattern: anger–happiness pairings showed robust facial dominance effects, anger–sadness pairings showed vocal dominance effects, and happiness–sadness pairings produced more heterogeneous impressions, including blended responses and third-category interpretations (i.e., anger or disgust for sad face–happy voice pairings). Together, these findings challenge a single-modality dominance view and suggest that bimodal emotion decoding depends on the specific emotions being integrated. To explain these pairing-dependent patterns, we propose and provide initial support for an emotion-similarity account, in which the relative influence of faces and voices depends on how distinguishable the competing emotions are within each modality.

Emotions are central to social life. In everyday interactions, people constantly infer how others feel and use these inferences to guide conversation, coordinate behavior, and decide when to offer support or keep their distance (Niedenthal & Brauer, 2012; Van Kleef, 2009; Van Kleef & Côté, 2022). These judgments rely heavily on nonverbal cues. People read emotions not only from the face, but also from the voice—its pitch, intensity, and rhythm (Elfenbein & Ambady, 2002; Laukka & Elfenbein, 2021; Sauter et al., 2010; Scherer et al., 2001). In many situations, the information provided by these modalities converges and points to the same emotion, making decoding relatively straightforward. However, emotional signals can also be mixed, regulated, or masked, and facial and vocal expressions do not always communicate the same emotion. It is precisely in these moments of conflict that misunderstanding, miscoordination, and social friction are most likely to arise, making it especially important to understand how people interpret discrepant emotional signals.

Consider, for example, the following scenario. After a negative event, a colleague responds to your concern with a faint smile while speaking in a sad tone. How do you interpret these mixed cues? Do you rely more on the facial expression and infer happiness, rely more on the vocal expression and infer sadness, or integrate both cues into a blended impression (e.g., simultaneous happiness and sadness)? This example illustrates a central question in audiovisual emotion perception: how people integrate facial and vocal cues when the two modalities signal different emotions.

1. Integration of emotional information from faces and voices

A growing literature examines how people integrate facial and vocal cues during emotion perception (e.g., Chen et al., 2024; de Gelder & Vroomen, 2000; Gil et al., 2016; Liu et al., 2021; Massaro & Egan, 1996; Vroomen et al., 2001). In an early study, Massaro and Egan (1996) paired computer-generated facial expressions (happiness, anger, and neutral) with a single spoken word in a happy, angry, or

* Corresponding author at: Department of Psychology and Behavioral Sciences, Zhejiang University, 866 Yuhangtang Rd, Hangzhou, Zhejiang Province, PR China.
E-mail address: x.fang@zju.edu.cn (X. Fang).

https://doi.org/10.1016/j.jesp.2026.105015
Received 11 May 2026; Received in revised form 28 September 2026; Accepted 3 October 2026
Available online 9 October 2026
0022-1031/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

neutral tone. Participants were asked to categorize targets as happy or angry. Although judgments reflected information from both modalities, facial cues exerted a stronger influence—an effect they termed facial dominance. Using more naturalistic materials, Collignon et al. (2008) paired dynamic facial expressions with nonverbal emotional vocalizations and similarly found that observers tended to rely more on facial information when the face and voice conveyed different emotions. Converging findings from related paradigms reported similar patterns (de Gelder et al., 2000; Gil et al., 2016; Watson et al., 2014).

More recently, Kawahara et al. (2021) demonstrated the generality of this pattern across different developmental stages and cultures. Specifically, participants in Japan and the Netherlands, ages 5–6, 11–12, and 18–32 years, viewed congruent (angry face–angry voice and happy face–happy voice) and incongruent (angry face–happy voice and happy face–angry voice) pairings using dynamic facial expressions with synchronized lip movements and nonverbal emotional vocalizations. On incongruent trials, participants chose the emotion based on the facial cue (e.g., anger for an angry face–happy voice) in over 75% of the cases across all age groups and both cultures, underscoring the robustness of facial dominance in the perception of audiovisual emotion signals.

Despite consistent support for facial dominance, several features of this prior work may constrain the generalizability of these findings. Most notably, many studies have relied on binary forced-choice judgments that restrict responses to the emotion conveyed by the face or by the voice (de Gelder & Vroomen, 2000; Kawahara et al., 2021; Watson et al., 2014). Because this format demands a single category choice, it cannot capture (a) blended impressions, in which the overall expression is perceived as containing emotions signaled by both the face and the voice, or (b) emergent impressions, in which the combination of facial and vocal cues gives rise to an overall interpretation that is not simply related to the emotion conveyed by either modality alone. Yet perceivers can represent multiple emotions at different intensities within a single facial expression (Fang et al., 2018, 2019; Fang & Kawakami, 2025), and cross-modal conflict may yield a qualitatively different percept, similar to fusion effects in audiovisual speech (McGurk & MacDonald, 1976).

A second limitation concerns the comparability of evidence across modalities. Many previous studies have not always assessed or equated the perceived emotional intensity of facial and vocal cues. Because facial expressions often convey more intense emotional signals than vocal expressions (see a meta-analysis by Elfenbein & Ambady, 2002), an apparent facial advantage may partly reflect differences in signal intensity rather than a modality-specific bias. Related evidence suggests that facial dominance is flexible rather than fixed. When noise made one channel less clear, observers relied more on the other, although adding noise to the face impaired judgments more than adding noise to the voice (Collignon et al., 2008).

A third limitation of past work is that ecological validity has been limited by the imperfect coupling between articulation and facial movement. Much of the past research has used dynamic facial displays without speech-synchronized lip movements (e.g., Collignon et al., 2008). However, in everyday conversation, articulation changes the lower-face musculature and the visible configuration of the mouth (Ekman et al., 2002; Fang et al., 2022), Notably, research on audiovisual speech perception suggests that dynamic, articulated faces can substantially shape integration (Campanella & Belin, 2007; McGurk & MacDonald, 1976), raising the possibility that nonsynchronized stimuli may misrepresent the relative influence of faces and voices in emotion perception.

2. Effects of emotion pairings on facial–vocal emotion integration

Beyond these methodological constraints, prior research has examined several emotion combinations, although anger–happiness conflicts have been particularly prominent. Relatively little work, however, has directly tested whether the relative influence of facial and vocal cues varies across specific emotion pairings. This leaves open whether facial dominance generalizes across emotion pairings or depends on the particular emotions in conflict. Research on audiovisual speech perception offers a useful example of how a specific combination of conflicting signals can shape the outcome of integration. In the classic McGurk paradigm (McGurk & MacDonald, 1976), when visual lip movements (e.g., /ga/) conflict with the auditory syllable (e.g., /ba/), observers typically report hearing a third, fused syllable (e.g., /da/). This fusion is commonly interpreted as an integration of both inputs. The visual articulation for /ga/ resembles /da/, and the acoustic evidence for /ba/ also overlaps with /da/, making /da/ a plausible compromise supported by both sources. Applying this logic to emotion perception, when facial and vocal cues conflict, perceivers may likewise seek an overall interpretation that is reasonably compatible with both modalities.

One factor that may shape this integration is the similarity between the competing emotions within each modality. Here, similarity refers to the extent to which emotion categories share expressive features within a given modality and are therefore more difficult to distinguish. Because faces and voices convey emotion through different expressive features, the similarity relations among emotion categories may differ across modalities. For facial expressions of basic emotions, this similarity structure appears to be organized largely by valence. Negative expressions (e.g., anger, sadness, fear, disgust) share more features with one another than with positive expressions (e.g., happiness), which may make negative faces more likely to be confused with one another (Cordaro et al., 2018; Ekman et al., 2002; Pan et al., 2026; Susskind et al., 2007). For vocal expressions, similarity appears to be organized more by arousal. High-arousal emotions (e.g., anger, fear, disgust, high-arousal happiness) tend to share acoustic features, such as high pitch, fast speech rate, and high intensity, whereas low-arousal emotions (e.g., sadness) are typically characterized by low pitch, low pitch variability, slow speech rate, and low vocal intensity, and therefore occupy acoustic spaces that are more distinct from those of high-arousal emotions (Bachorowski & Owren, 1995; Goudbeek & Scherer, 2010; Laukka et al., 2005; Laukka et al., 2016; Pell et al., 2009; Scherer, 2018). Thus, the relative distinctiveness of two competing emotions can differ between faces and voices.

When two emotions are more distinct in facial than in vocal expressions, facial cues should exert greater influence on perception; when they are more distinct in vocal than in facial expressions, vocal cues should exert greater influence. When the two emotions are well differentiated in both modalities, the emotions depicted in the face and voice may play more comparable roles in overall perception. We refer to this perspective as an emotion-similarity account of audiovisual integration. Importantly, similarity here concerns relations among emotion categories within a modality rather than the overall clarity of an individual signal. Even clearly recognized expressions can be more similar to some emotions than to others (e.g., a sad face may be clearly perceived as sad while still sharing more facial features with anger than with happiness), and these similarity relations may shape which interpretation becomes most plausible when facial and vocal cues conflict.

Several lines of prior work provide indirect support for this possibility. Takagi et al. (2015), using all six basic emotions, found that facial–vocal influence varied across emotion categories and attentional conditions, although their analyses did not directly compare specific emotion pairings. Related work on face–body integration shows that contextual influence can vary with the perceptual similarity between emotions (Aviezer et al., 2008; Aviezer et al., 2011; Lecker et al., 2020). A recent systematic review further identified both dimensional proximity and shared perceptual features (i.e., emotion seeds) as potential determinants of face–body incongruence effects (Puffet et al., 2025). Together, these findings suggest that the particular emotions being combined can shape integration, but whether this also affects facial–vocal weighting remains unclear.

3. The present research

In the present research, we sought to examine how perceivers integrate facial and vocal cues when the two modalities convey different emotions. We focused on three incongruent pairings. Anger–happiness pairings combine opposite-valence emotions that tend to be far apart in facial configuration yet relatively similar in high-arousal vocal space. Anger–sadness pairings, alternatively, combine two negative emotions that are closer in facial space but strongly diverge in arousal in vocal space. Finally, happiness–sadness pairings differ on both valence and arousal, with the two signals being relatively distant in both modalities. Together, these contrasts provided us with the opportunity to test whether facial dominance extends beyond anger–happiness contrasts and, more broadly, whether the relative influence of facial and vocal cues varies systematically across emotion pairings.

To address key methodological limitations of prior work, we created dynamic, articulated facial expressions with speech-synchronized lip movements, paired them with intensity-matched vocalizations across modalities, and used response formats that did not force a single categorical choice. Across six preregistered laboratory experiments, we tested these pairings using two complementary measurement approaches. In Experiments 1–3, participants provided multidimensional ratings of the expresser's overall emotion, which allowed them to report blends rather than choose a single label. In Experiments 4–6, to further reduce response constraints, participants offered open-ended descriptions of the emotions, potentially capturing a wider range of perceptions.

4. Open practices

In accordance with current open practice recommendations, we reported how we determined our sample sizes, all data exclusions, all manipulations, and all measures for each experiment. The experimental designs and analysis plans were preregistered before data collection. The preregistrations for Experiments 1–3 and Experiments 4–6 are available at https://aspredicted.org/74C_8BT and https://aspredicted.org/PBG_KNV, respectively. Deviations from the preregistered analysis plans are summarized in Section S1 of the Supplementary Materials. Data, analysis code, experiment program, and shareable stimulus materials are available at https://osf.io/mw5uq/?view_only=025f27553962443c8e85da8f75a226b3. Stimulus materials containing identifiable faces are posted publicly only when models provided consent; other materials are available from the corresponding author upon reasonable request. Analyses were conducted in R version 4.3.3 (R Core Team, 2024) using the package bruceR (Version 2024.6; Bao, 2024), emmeans (Version 1.11.2; Lenth, 2025), ggplot2 (Version 3.5.2; Wickham, 2016), ggpubr (Version 0.6.1; Kassambara, 2025), lme4 (Version 1.1.37; Bates et al., 2015), lmerTest (Version 3.1.3; Kuznetsova et al., 2017), and rstatix (Version 0.7.2; Kassambara, 2023).

5. Experiments 1–3: Multidimensional ratings

In Experiments 1–3, participants viewed short videos depicting models producing facial expressions while speaking. After each video, they rated the expresser's overall emotional state on five dimensions—anger, disgust, fear, happiness, and sadness. Experiment 1 examined judgments for anger–happiness pairings (angry face–happy voice, happy face–angry voice), Experiment 2 examined judgments for anger–sadness pairings (angry face–sad voice, sad face–angry voice), and Experiment 3 examined judgments for happiness–sadness pairings (happy face–sad voice, sad face–happy voice).

5.1. Method

5.1.1. Participants

For each experiment, the target sample size was preregistered at 60 participants based on an a priori power analysis in GPower 3.1 (Faul et al., 2007), using the ANOVA: Repeated measures, within factors procedure for the focal 2 × 2 Incongruent Configuration × Rating Scale interaction. The four cells were entered as four repeated measurements, which we note does not exactly represent the focal interaction in GPower. Assuming $\eta_p^2 = 0.06$, $\alpha = 0.05$, and 80% power, this analysis yielded a target of 60 participants per experiment.

Participants were Chinese university students who received either 15 RMB or course credit equivalent to 0.5 h of research participation. Demographic characteristics are reported in Table 1. All procedures were approved by the university's ethics committee, and all participants provided written informed consent. Because the final analyses included all five rating scales, we conducted sensitivity analyses for the corresponding 2 × 5 interaction using five measurements. With $N = 60$, $\alpha = 0.05$, and 80% power, and using the Greenhouse–Geisser ε values from the corresponding ANOVAs (ε = 0.400, 0.508, and 0.512 for Experiments 1–3, respectively), the minimum detectable effects were Cohen's f = 0.313, 0.288, and 0.287 ($\eta p^2 = 0.089$, 0.077, and 0.076), respectively.

5.1.2. Development and validation of bimodal stimuli

We constructed a stimulus set in which emotional intensity was matched across faces and voices in three steps (see Section S2 of the Supplementary Materials for details). First, we pretested 15 Chinese sentences that were neutral in semantic content and similar in length. Thirteen participants rated the valence of each sentence, and we selected six sentences with neutral valence (M = 0.01, SD = 0.21) on a scale where 0 indicated neutral valence.

Second, we recruited six models (three men and three women) to produce dynamic facial and vocal expressions for each sentence in three emotions: anger, happiness, and sadness. A research assistant trained in the Facial Action Coding System (FACS; Ekman et al., 2002) monitored the recording sessions to promote clear and consistent expressions. To increase the pool of candidate stimuli, each sentence was performed at least three times per emotion and per modality, yielding 961 raw clips in total.

Third, given the large number of original recordings, the second author first screened the materials and selected candidate clips with moderate intensity of the intended emotion. Twenty-eight participants then rated the perceived intensity of the intended emotion in each clip from 0 (“not at all”) to 100 (“very much”). Using these ratings, we created incongruent face–voice pairings with approximately matched perceived intensity of the intended emotion across modalities (see Table S2). Each model–sentence combination was retained only if both directional mismatches within a pairing met this criterion. For example, in the anger–happiness pairing, both happy face–angry voice and angry face–happy voice stimuli had to show comparable perceived intensity between the face and the voice.

Table 1
Participant Information for Experiments 1–6.

Evaluation Format	Experiments	N	Mean Age (SD)	Gender Distribution
Multidimensional Ratings	Experiment 1 (Anger–Happiness)	60	21.57 (2.32)	21 m, 39 f
	Experiment 2 (Anger–Sadness)	60	21.55 (2.60)	18 m, 42 f
	Experiment 3 (Happiness–Sadness)	60	21.62 (2.24)	17 m, 43 f
Open-ended Responses	Experiment 4 (Anger–Happiness)	60	20.67 (2.58)	27 m, 33 f
	Experiment 5 (Anger–Sadness)	59	21.85 (3.08)	15 m, 44 f
	Experiment 6 (Happiness–Sadness)	60	20.93 (2.24)	22 m, 38 f

Note. The data from one participant in Experiment 5 who failed to finish the experiment was excluded from the analyses. Under gender distribution, “m” denotes male and “f” denotes female.

This procedure yielded 24 incongruent stimuli per experiment (4 models × 2 incongruent configurations × 3 neutral sentences). The corresponding facial and vocal clips were also combined to form 24 congruent stimuli per experiment (4 models × 2 congruent configurations × 3 neutral sentences). The specific models and sentences used in each experiment are listed in Table 2. Although the incongruent stimuli were the focus of the current investigation, to limit participants focusing on incongruency and applying a deliberate response strategy, congruent stimuli were also included.

5.1.3. Procedure

The experiment was programmed in Psychopy 2021.2.3 (Peirce et al., 2019). Participants completed the task individually in a quiet room while wearing headphones, with the volume adjusted to a comfortable level. Visual stimuli were presented at a viewing distance of approximately 60 cm and subtended approximately 12° × 6.75° of visual angle. Before each session, participants were instructed: "Please evaluate the emotions expressed by the individual in the video [or audio]. On a scale from 0 ('not at all') to 100 ('very much'), rate the extent to which the individual expressed anger, disgust, fear, happiness, and sadness." Each trial began with a one-second fixation cross ("+"), followed by the presentation of the stimulus. Once the stimulus disappeared, the five emotion rating scales were presented. All five emotion ratings used the same response format and endpoint anchors, and their order was randomized across participants. There was no time limit to complete the ratings and participants were instructed to press the spacebar to continue to the next trial, which was preceded by a one-second blank screen.

The experiment consisted of three sessions. The first session evaluated bimodal (congruent and incongruent) emotional expressions, and the subsequent two sessions assessed unimodal (visual and auditory) emotional expressions (see Fig. 1A). The unimodal sessions were kept separate to provide relatively independent assessments of the component facial and vocal expressions while minimizing carryover from bimodal judgments and frequent switching between stimulus formats. The bimodal session comprised two 48-trial blocks, with each stimulus presented once per block in randomized order. Responses were averaged across blocks to improve the stability of the estimates. Cross-block reliability was reasonably high in all three experiments (ICCs = 0.72–0.79), and Block did not modulate the critical Incongruent Configuration × Rating Scale interaction, $ps \geq 0.224$ (see Section S3 of the Supplementary Materials). The visual and auditory unimodal sessions each consisted of 24 trials (4 models × 2 emotions × 3 neutral sentences), presented in a random order. The order of the two unimodal sessions was counterbalanced across participants.

Table 2
Models and sentences used in each experiment.

Experiments	Models	Sentences	Duration
Experiment 1 (Anger–Happiness)	m1, f1	I just got to the station. Please wait a moment. Your stuff is over there.	1.10 s - 1.70 s
	m2, f2	I just got to the station. I'll be there in a moment. Let me tell you something.	
Experiment 2 (Anger–Sadness)	m1, f1	I just got to the station. My bag is at home. Let me tell you something.	1.24 s - 2.18 s
	m2, f2	Please wait a moment. I'll be there in a moment. Let me tell you something.	
Experiment 3 (Happiness–Sadness)	m2, f1	I just got to the station. Please wait a moment. My bag is at home.	1.12 s - 2.18 s
	m1, f2	I just got to the station. I'll be there in a moment. Let me tell you something.	

Note. In the "Models" column, "m" denotes male, and "f" denotes female. The accompanying number is the assigned model identifier.

5.2. Results

5.2.1. Validation of unimodal stimuli

To verify that the unimodal (visual or auditory) stimuli conveyed the intended emotions, we conducted separate one-way repeated-measures ANOVAs for each unimodal stimulus type, with Rating Scale (anger, disgust, fear, happiness, and sadness) as the within-subject factor. Because the assumption of sphericity was violated in all analyses, Greenhouse–Geisser corrections were applied throughout. We then conducted four paired-samples *t*-tests for each stimulus type, comparing the intended-emotion rating with each of the four nonintended emotion ratings, with Holm adjustment applied across the four comparisons. The effect of Rating Scale was significant for all stimulus types ($ps < 0.001$, and the intended emotion was rated more intensely than each nonintended emotion in all comparisons (all Holm-adjusted $ps \leq 0.012$). Overall, these findings indicate that the unimodal stimuli successfully conveyed their intended emotions. Full ANOVA results are reported in Table S4, and the corresponding Holm-adjusted comparisons are reported in Table S5.

5.2.2. Incongruent bimodal stimuli

Because our central question concerned how participants interpreted conflicting facial and vocal signals, we focused the main analyses on the incongruent conditions. *Descriptive statistics for congruent and incongruent* configurations *are reported in Table S6 and Table S7, respectively. For each experiment, we conducted a repeated-measures ANOVA on emotion ratings, with* Incongruent Configuration *and Rating Scale as within-subjects factors*. Incongruent Configuration contrasted angry face–happy voice with happy face–angry voice in Experiment 1, angry face–sad voice with sad face–angry voice in Experiment 2, and happy face–sad voice with sad face–happy voice in Experiment 3. Rating Scale included happiness, anger, sadness, disgust, and fear. Greenhouse–Geisser corrections were applied when sphericity was violated. Key follow-up comparisons are reported below, with the full pairwise comparisons in Table S8.

Experiment 1: Anger–Happiness. The main effects of Incongruent Configuration, $F(1, 59) = 10.09$, $p = .002$, $\eta_p^2 = 0.146$, 90% CI [0.034, 0.287], Rating Scale, $F(2.93, 172.62) = 138.49$, $p < .001$, $\eta_p^2 = 0.701$, 90% CI [0.643, 0.746], and their interaction, $F(1.60, 94.45) = 177.04$, $p < .001$, $\eta_p^2 = 0.750$, 90% CI [0.681, 0.799], were all significant. Simple effects analyses of the interaction revealed a facial dominance effect. As depicted in Fig. 2, when angry faces were paired with happy voices, anger intensity ratings did not differ from disgust intensity ratings, $t(59) = 1.14$, $p = .260$, Cohen's $d = 0.15$, 95% CI [−0.11, 0.41], but they were higher than fear, sadness, and happiness intensity ratings, $ps < 0.001$. In contrast, when happy faces were paired with angry voices, happiness intensity ratings were significantly higher than intensity ratings of all other emotions, $ps < 0.001$.

Although the relatively high disgust intensity ratings in the angry face–happy voice condition were not predicted, they likely reflect the fact that facial expressions of anger and disgust are often confused, especially among East Asian perceivers (Fang et al., 2019; Fang & Kawakami, 2024; Jack et al., 2009). In the unimodal validation, angry faces were rated primarily as anger but also elicited relatively elevated disgust intensity ratings, consistent with partial perceptual overlap between anger and disgust (Table S5). This perceptual overlap may have contributed to the similar anger and disgust intensity ratings in the bimodal condition. Together, this pattern of results primarily provides evidence for a facial dominance effect in anger–happiness conflicts, with responses more closely aligned with the emotional meaning of the face than with that of the voice.

Experiment 2: Anger–Sadness. The main effects of Incongruent Configuration, $F(1, 59) = 23.59$, $p < .001$, $\eta_p^2 = 0.286$, 90% CI [0.135, 0.428], Rating Scale, $F(2.89, 170.62) = 113.59$, $p < .001$, $\eta_p^2 = 0.658$, 90% CI [0.592, 0.708], and their interaction, $F(2.03, 119.89) = 139.32$,

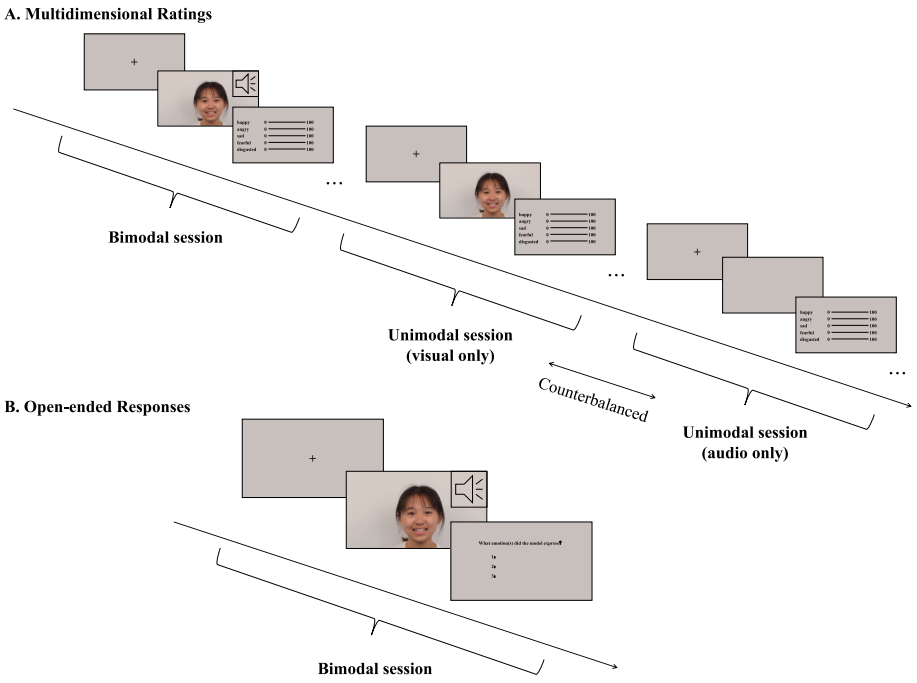


Fig. 1. Overview of the Procedure in Experiments 1–6.
Note. Panel A shows the multidimensional-rating task used in Experiments 1–3. Panel B shows the open-ended-response task used in Experiments 4–6.

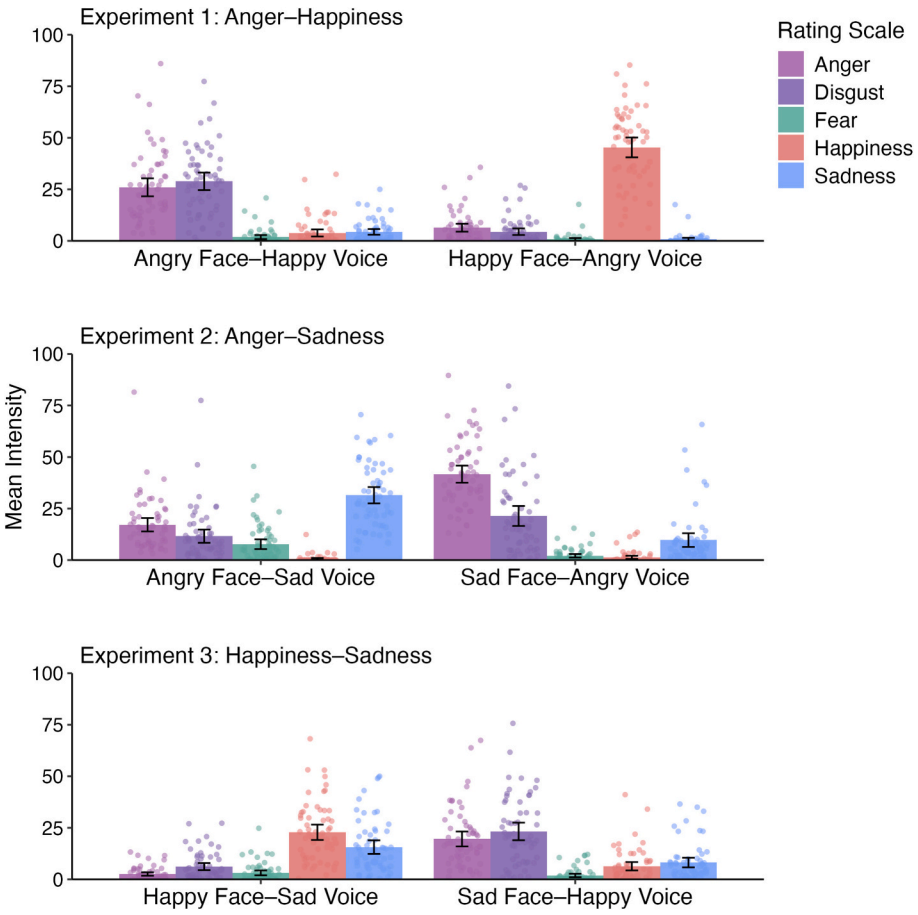


Fig. 2. Emotion Ratings of Incongruent Face–Voice Pairings in Experiments 1–3.
Note. Error bars indicate 95% confidence intervals.

$p < .001$, $\eta_p^2 = 0.702$, 90% CI [0.632, 0.755], were significant. Simple effects analyses of the interaction indicated that when angry faces were

paired with sad voices, sadness intensity ratings were significantly higher than anger, disgust, fear, and happiness intensity ratings, $ps < 0.001$. When sad faces were paired with angry voices, anger intensity ratings were significantly higher than the intensity ratings of all other emotions, $ps < 0.001$. Together this pattern of findings provides evidence for a vocal dominance effect for anger–sadness conflicts.

Experiment 3: Happiness–Sadness. The main effects of Incongruent Configuration, $F(1, 59) = 14.05$, $p < .001$, $\eta_p^2 = 0.192$, 90% CI [0.063, 0.337], Rating Scale, $F(3.32, 195.97) = 34.39$, $p < .001$, $\eta_p^2 = 0.368$, 90% CI [0.278, 0.441], and their interaction, $F(2.05, 120.86) = 70.95$, $p < .001$, $\eta_p^2 = 0.546$, 90% CI [0.448, 0.621], were significant. Simple effects analyses of the interaction indicated that when happy faces were paired with sad voices, intensity ratings of happiness and sadness did not differ after Bonferroni correction, $t(59) = 2.74$, $p = .008$, Cohen's $d = 0.35$, 95% CI [0.10, 0.61], and exceeded intensity ratings of anger, disgust, and fear, $ps < 0.001$. This pattern suggests a blended impression in which perceptions contain emotions signaled by both the face and the voice. In contrast, when sad faces were paired with happy voices, intensity ratings of anger and disgust did not differ after correction, $t(59) = 2.10$, $p = .040$, Cohen's $d = 0.27$, 95% CI [0.01, 0.53], and were higher than intensity ratings of fear, happiness, and sadness, $ps < 0.001$. This pattern of findings suggests an emergent impression in which perceptions are not simply related to emotions conveyed by either modality alone but related to alternative emotions.

5.2.3. Robustness to residual intensity differences

Although our stimuli were matched in perceived intensity at the stimulus level, participants could still experience residual intensity differences between the face and the voice on individual trials. To test whether these differences could account for the pairing-dependent effects, we fitted separate linear mixed-effects models for each experiment predicting trial-level bimodal emotion intensity ratings from Incongruent Configuration, Rating Scale, their interaction, and the corresponding participant-specific face–voice intensity difference. The models included random intercepts for participants and video stimuli and a by-participant random slope for Incongruent Configuration; full model-fitting details are reported in Section S5 of the Supplementary Materials. The models reproduced the overall ANOVA pattern, with a facial dominance effect for anger–happiness pairings, a vocal dominance effect for anger–sadness pairings, and a nondominance effect for happiness–sadness pairings. Although some follow-up comparisons differed from the ANOVAs, the overall pairing-dependent integration pattern remained after controlling for participant-specific differences in perceived face–voice intensity (see Section S5).

5.2.4. Exploratory similarity analyses

After establishing the main patterns of ratings and their robustness to residual intensity differences, we conducted two exploratory analyses to examine whether these patterns were consistent with the proposed emotion-similarity account. First, we examined whether the stimuli showed the assumed similarity structure within each modality. Using facial features extracted with OpenFace (Baltrusaitis et al., 2018) and acoustic features extracted with OpenSMILE (Eyben et al., 2010), we quantified the physical similarity among anger, happiness, and sadness separately for faces and voices. Full details of this analysis are reported in Section S6 of the Supplementary Materials. The results broadly supported the assumed structure: anger and sadness were relatively closer in facial space, whereas anger and happiness were relatively closer in vocal space.

Second, we tested whether bimodal judgments leaned toward the modality in which the two competing emotions were more perceptually distinct. For each incongruent stimulus, we used the corresponding unimodal stimuli produced by the same actor uttering the same sentence to compare the two emotions separately within the facial and vocal modalities. For example, for an angry face paired with a happy voice, we compared the angry face with the happy face, as well as the happy voice with the angry voice. Each unimodal stimulus was represented by its average ratings across participants on the five emotion scales, and the Euclidean distance between two rating profiles indexed the perceptual distinctiveness of the competing emotions within that modality. We then calculated relative distinctiveness as facial distance minus vocal distance, with larger values indicating greater distinctiveness in faces than in voices. Modality dominance was calculated as the bimodal rating of the facial emotion minus that of the vocal emotion, with larger values indicating greater facial dominance. Across the 72 incongruent stimuli from Experiments 1–3, relative distinctiveness was positively correlated with modality dominance, $r(70) = 0.58$, $p < .001$. Thus, bimodal judgments leaned toward the modality in which the competing emotions were more perceptually distinct.

Together, these exploratory analyses suggest that the patterns observed across emotion pairings in Experiments 1–3 were consistent with the assumed similarity structure and were systematically related to how clearly each modality distinguished the competing emotions.

6. Experiments 4–6: Open-ended responses

Building on the findings from Experiments 1–3, we examined whether the distinct patterns observed across emotion pairings would emerge when participants were not restricted to predefined emotion scales. In Experiments 4–6, participants described the perceived emotions in their own words, providing us with the ability to test whether similar patterns would emerge under a less constrained response format. The designs mirrored Experiments 1–3, with Experiment 4 focusing on anger–happiness pairings, Experiment 5 focusing on anger–sadness pairings, and Experiment 6 focusing on happiness–sadness pairings.

6.1. Method

6.1.1. Participants

In accordance with Experiments 1–3, we aimed to recruit 60 participants per experiment. The final samples included 60, 59, and 60 Chinese university students in Experiments 4–6, respectively, who received a monetary compensation of 15 RMB or course credit equivalent to 0.5 h of research participation. Demographic characteristics are reported in Table 1. The university ethics committee approved all procedures and written informed consent was obtained from all participants. Sensitivity analyses in G*Power showed that the final samples could detect effects of approximately Cohen's $d_z = 0.37$ for two-tailed paired-samples t-tests comparing the two incongruent configurations on weighted means related to valence or arousal, with $\alpha = 0.05$ and power = 0.80.

6.1.2. Procedure

The procedure was similar to Experiments 1–3, with one key difference being that only the bimodal session was administered. Each experiment comprised one block of 48 bimodal stimuli (see Fig. 1B) and began with a one-second fixation cross ("+"), followed by the stimulus. After the video ended and was removed from the screen, three blank text boxes appeared. Participants were instructed to type the emotions expressed by the model, entering at least one and no more than three emotions. Responses were typed via keyboard with no time limit. Upon completing their responses, participants pressed the "NEXT" button to proceed to the subsequent trial, with a one-second blank screen between trials.

6.2. Results

6.2.1. Open-ended response coding

Open-ended responses were coded into emotion categories by first compiling terms that appeared more than once into a candidate list for manual review. Two trained research assistants independently classified each candidate term as emotion related or nonemotion related, with

discrepancies resolved by the corresponding author. Interrater reliability for this initial classification was calculated before adjudication and showed moderate agreement, Cohen's $\kappa = 0.50$, with a raw agreement rate of 79.83%. Emotion-related candidate terms were then consolidated by merging close synonyms. This synonym-merging step was conducted by the two original coders and an additional trained assistant. Categorizations showed high agreement among the three coders before adjudication, Fleiss' $\kappa = 0.86$, with 80.35% of terms receiving identical category assignments across all three coders. After adjudication, this procedure yielded 43 distinct emotion categories. Remaining low-frequency emotion terms were then assigned to one of these categories when possible or coded as "other." For full coding details see Section S7 in the Supplementary Materials.

Of the 43 categories, 40 were mapped onto valence-arousal coordinates using normative ratings from the Chinese EmoBank (Lee et al., 2022). EmoBank norms were not available for the remaining three categories (i.e., *neutral*, *forced smile*, and *sajiao*). For each incongruent configuration, we calculated frequency-weighted mean valence and arousal, along with the proportions of positive, negative, and neutral responses based on whether category valence was above, below, or equal to 5, the neutral midpoint of the EmoBank valence scale. We also compared the two incongruent configurations within each experiment at the participant level on mean valence and mean arousal.

Experiment 4: Anger–Happiness. Because happiness and anger differ primarily in valence but are both relatively high in arousal, valence better indicates whether responses aligned more closely with the facial or vocal cue. For happy face–angry voice pairings, the weighted mean valence was 6.39 (+1.39 relative to neutral), and 83.2% of mapped responses were positive. For angry face–happy voice pairings, the weighted mean valence was 3.52 (−1.48 relative to neutral), and 88.4% of responses were negative (see the red pairings in Fig. 3). Participant-level analyses showed that mean valence differed significantly across the two configurations, $t(59) = 24.79$, $p < .001$, Cohen's $d = 3.20$, 95% CI [2.942, 3.459], whereas the difference in mean arousal was not significant, $t(59) = -0.70$, $p = .484$, Cohen's $d = -0.091$, 95% CI [−0.349, 0.167]. Together, these results suggest that, even when participants generated emotion labels freely, responses aligned more closely with the valence of the facial cues, consistent with the facial dominance effect observed for anger–happiness pairings in Experiment 1.

Experiment 5: Anger–Sadness. Because anger and sadness are both negative in valence but differ clearly in arousal, arousal better indicates whether responses aligned more closely with the facial or vocal cue. The weighted mean arousal was 5.23 for angry face–sad voice pairings and 6.25 for sad face–angry voice pairings (see the blue pairings in Fig. 3). Participant-level analyses showed that mean arousal differed significantly across the two configurations, $t(58) = 10.69$, $p < .001$, Cohen's $d = 1.39$, 95% CI [1.132, 1.653], whereas mean valence showed little difference, $t(58) = -1.89$, $p = .064$, Cohen's $d = -0.25$, 95% CI [−0.507, 0.015]. Together, these results suggest that, even when emotion labels are generated freely, responses aligned more closely with the arousal of the vocal cues, consistent with the vocal dominance effect for anger–sadness pairings in Experiment 2.

Experiment 6: Happiness–Sadness. Although happiness and sadness differ in both valence and arousal, they are more strongly differentiated by valence. Notably, the two incongruent configurations yielded different impressions rather than a single pattern of modality dominance. For happy face–sad voice pairings, the weighted mean valence was 4.99 (−0.01 relative to neutral), and the valence distribution was nearly balanced, with 47.8% positive versus 51.9% negative, indicating relatively mixed emotion perceptions. In contrast, for sad face–happy voice pairings, the weighted mean valence was 3.63 (−1.37 relative to neutral), with a mean arousal of 5.96, and the distribution was strongly negative (85.5% negative vs. 14.2% positive), indicating predominantly negative emotion perceptions (see the green pairings in Fig. 3). Participant-level analyses showed that mean valence differed significantly across the two configurations, $t(59) = 9.11$, $p < .001$, Cohen's $d = 1.18$, 95% CI [0.918, 1.435]. Mean arousal also differed significantly, although the effect was smaller, $t(59) = 5.53$, $p < .001$, Cohen's $d = 0.71$, 95% CI [0.455, 0.972]. Together, these results suggest that happiness–sadness pairings are better characterized by different response profiles across configurations than by dominance of a single modality. This pattern was broadly consistent with that observed for happiness–sadness pairings in Experiment 3.

Together, Experiments 4–6 provided qualitative convergence with the pairing-dependent patterns observed in Experiments 1–3 under a less constrained response format. Even with open-ended responses, the three emotion pairings produced integration patterns broadly similar to those observed with multidimensional ratings.

7. General discussion

Across six experiments using multidimensional ratings and open-ended response formats, we found that the perception of incongruent facial and vocal emotions varied systematically by emotion pairing. Anger-happiness pairings showed a facial dominance pattern, with responses more strongly aligned with facial than vocal signals. Anger-sadness pairings showed a vocal dominance pattern, with responses

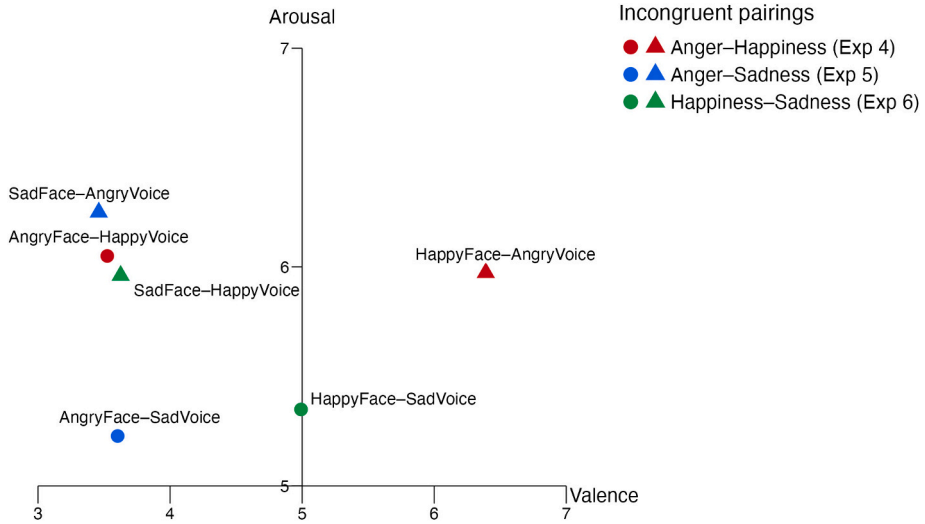


Fig. 3. Valence–Arousal Ratings of Open-Ended Responses for Incongruent Face–Voice Pairings in Experiments 4–6.

more strongly aligned with vocal than facial signals. Happiness-sadness pairings yielded more heterogeneous impressions. Happy faces paired with sad voices produced a blend of happiness and sadness, whereas sad faces paired with happy voices elicited alternative negative categories, particularly anger and disgust. Taken together, these results challenge the idea of a general, context-free facial dominance effect in audiovisual emotion perception and instead point to pairing-dependent integration patterns.

These findings also have broader theoretical implications for how audiovisual emotion perception is conceptualized. Rather than asking whether faces or voices dominate in general, they suggest that modality influence depends on the specific emotions brought into conflict. When we describe some patterns as facial dominance or vocal dominance, we are referring to the direction in which participants' judgments are aligned, rather than to a claim that perceivers treat facial and vocal signals as independent cues. This interpretation is compatible with holistic and Gestalt perspectives on multimodal communication, in which signals from different channels are organized into an overall multimodal impression rather than reduced to either channel alone (Holler & Levinson, 2019; Trujillo & Holler, 2023). Evidence for such integrative impressions was clearest in the happiness–sadness pairings, where conflicting facial and vocal cues produced blended responses in happy face–sad voice configurations and shifted anger/disgust responses in sad face–happy voice configurations.

7.1. An Emotion-similarity account of pairing-dependent integration

We propose an emotion-similarity account of these pairing-dependent outcomes. The core idea is that faces and voices carry different similarity structures and that perceivers integrate the two channels into an interpretation that is plausible given both. When the competing emotions are more perceptually distinct in one modality than in the other, that modality should exert greater influence, not because it is inherently more reliable, but because it more strongly constrains the plausible interpretations of that particular conflict. This account extends research on emotion seeds and face–body integration, which shows that partial similarities among expressive signals can shape contextual emotion perception (Aviezer et al., 2008; Aviezer et al., 2011; Lecker et al., 2020; Puffet et al., 2025). Our findings suggest that modality-specific similarity may likewise shape the relative influence of facial and vocal cues when they conflict.

The three pairings tested here differed in their predicted similarity structure. In anger–happiness pairings, anger and happiness are relatively distinct in facial space but less distinguishable in vocal space, consistent with the observed facial dominance pattern. In anger–sadness pairings, they are less distinguishable in facial space but more distinct in vocal space, consistent with the observed vocal dominance pattern. In happiness–sadness pairings, the two emotions are relatively distinct in both modalities. Under these conditions, neither channel clearly resolves the conflict, making blended or emergent impressions more likely. Blending was most evident when happy faces were paired with sad voices and responses retained evidence of both emotions. Emergent impressions were more evident when sad faces were paired with happy voices and responses shifted toward anger and disgust. Because sad faces share some facial features with anger and disgust, whereas happy voices share high-arousal acoustic features with these categories, both channels may provide partial support for these off-target interpretations. From this perspective, emergent impressions are not random noise but structured outcomes shaped by modality-specific similarity relations.

The exploratory similarity analyses were consistent with this interpretation. The physical similarity analysis showed that the stimuli broadly followed the proposed modality-specific structure, with anger and sadness relatively closer in facial space and anger and happiness relatively closer in vocal space. The perceptual distinctiveness analysis more directly linked this structure to the response patterns. Specifically, bimodal judgments leaned toward the modality in which the competing emotions were more clearly distinguishable. Thus, the pairing-dependent influence of faces and voices appears to reflect, at least in part, how emotion categories are organized within each modality.

Although this account emphasizes expressive-feature similarity as a bottom-up source of perceptual organization, conceptual knowledge may also shape perceived relations among emotion categories through top-down processes. Conceptual structure predicts the perceptual organization of facial expressions even after physical similarity is controlled (Brooks & Freeman, 2018) and has been linked to neural representations of both facial and vocal emotions (Gao et al., 2024). Perceptual distinctiveness may therefore reflect both expressive and conceptual similarity, although the present study did not directly assess conceptual similarity or its contribution to modality weighting.

7.2. Additional factors shaping pairing-dependent integration

The emotion-similarity account should be viewed as an organizing framework rather than a complete theory. Although facial and vocal signals provide modality-specific similarity constraints, perceived relations among emotion categories may vary across perceiver groups. Prior research suggests that some facial emotion categories, including anger and disgust as well as fear and surprise, are less clearly differentiated among East Asian than Western perceivers (Fang et al., 2019; Fang & Kawakami, 2024; Jack et al., 2009, 2016). Such culturally patterned differences in emotion confusability may reshape perceived similarity spaces and thereby modulate integration outcomes for specific face–voice pairings. Thus, the same face–voice pairing may not necessarily produce the same integration pattern across perceiver cultures.

Ecological plausibility and familiarity may also shape how face–voice conflicts are interpreted. Some incongruences may evoke familiar social situations, such as a smile masking sadness, whereas others may not readily support a coherent interpretation. These learned expectations may influence whether perceivers integrate both channels, infer a mixed or regulatory state, or rely more strongly on one modality, thereby contributing to differences across emotion pairings.

7.3. Implications for multimodal emotion inference

These findings also have implications for contexts in which emotions are inferred from multimodal signals. Researchers and affective-computing systems should be cautious about treating one modality as the default basis for inferring emotions, especially when facial and vocal cues are discrepant. Such discrepancies may reflect mixed affect, masking, regulation, or socially pragmatic communication, and their interpretation may depend on which specific emotions are involved. Systems that rely on a fixed modality hierarchy may therefore miss the pairing-dependent nature of audiovisual emotion perception. Accordingly, models and coding schemes for inferences related to multimodal emotions should allow conflicting signals to produce not only face dominant or voice dominant judgments, but also blended and emergent impressions.

7.4. Constraints on generality

Several limitations qualify the scope of our conclusions. First, the facial and vocal expressions included in this research were posed rather than spontaneous. This design allowed us to test modality influence under controlled conditions, using stimuli in which intended emotions could be systematically combined across faces and voices and approximately matched in emotional clarity and intensity. In everyday interactions, mixed, masked, and regulated emotions do occur, but rarely as a fully crossed set of anger–happiness–sadness combinations. Posed stimuli therefore served as a controlled test for the integration mechanism, analogous to audiovisual speech studies that use constructed syllable pairings to reveal integration principles. Future work, however, should test more ecologically common conflicts, such as polite smiles

over negative feelings or irritated tone over otherwise neutral faces.

Second, Experiments 1–3 and 4–6 used different response formats, limiting direct comparisons between the two sets. Experiments 1–3 assessed emotion intensity on five predefined emotion scales, whereas Experiments 4–6 elicited open-ended descriptions that were coded into emotion categories and mapped onto valence–arousal norms. Correspondence between the two sets should therefore be interpreted as qualitative convergence, whereas some differences may partly reflect their distinct response demands. Nevertheless, the open-ended responses captured meanings beyond the intended emotion categories, including anxiousness, impatience, forced smile, and *sajiao* (Supplementary Table S12), suggesting that face–voice conflicts can elicit broader affective or communicative interpretations. This pattern is consistent with views that emotional expressions guide social inference and coordinate interaction, rather than only index discrete emotion categories (Crivelli & Fridlund, 2018; Scarantino, 2017; Van Kleef, 2016). Future research should examine these meanings in more naturalistic face–voice conflicts.

Third, we examined only three pairings, anger–happiness, anger–sadness, and happiness–sadness, each in a separate experiment. Our analyses did not directly compare the magnitude of facial and vocal influence across pairings, and our conclusions therefore concern the distinct response patterns observed within the three pairings tested. The design also did not provide a fully crossed test of valence and arousal. While our set included both a high-arousal negative emotion (anger) and a low-arousal negative emotion (sadness), only one relatively high-arousal positive basic emotion, happiness, was included. This asymmetry partly reflects our focus on prototypical expressions of basic emotion, for which happiness is the primary positive category. Future studies could use a broader and more systematically crossed set of emotions, including pairs of high-arousal negative emotions such as anger–fear or anger–disgust, to examine the respective roles of emotion category, valence, arousal, and the perceptual similarity of the conflicting emotions within each modality. They could also test whether emotion similarity constrains integration across other expressive channels, including postures, gestures, and verbal content.

Finally, the current research was conducted within a single national and cultural context. Prior research suggests that cultural background shapes reliance on emotional cues, with East Asian individuals often showing greater reliance on auditory information and Western individuals often giving more weight to visual cues (Chen et al., 2022; Chen et al., 2024; Liu et al., 2015a, 2015b; Tanaka et al., 2010). Beyond weighting, culture may also shape perceptual similarity among expressions, further modulating cross-modal integration. If ecological frequency and familiarity also play a role, cultural differences in the prevalence and interpretation of specific pairings could likewise shift integration outcomes. Consequently, the patterns of facial and vocal influence observed among Chinese participants may not generalize across cultures. Cross-cultural replications that systematically vary expresser and perceiver cultures are therefore needed to establish the boundary conditions of these effects.

8. Conclusion

Although emotional signals across modalities often align, real-world interactions also include emotional signals that are conflicting. The present studies provide novel evidence that modality influence in audiovisual emotional perception is not fixed, but varies across specific emotion pairings, challenging the assumption of a consistent facial dominance effect. To account for these patterns, we proposed and found initial support for an emotion-similarity account. According to this account, integration of cues depends on modality-specific similarity structures and the extent to which facial and vocal cues support a common interpretation. Although future work should provide more direct tests of this account and clarify how other factors, such as ecological plausibility and culture, shape integration outcomes, the present findings provide consistent evidence for the importance of examining specific cross-modal emotion pairings and their similarity structures.

CRediT authorship contribution statement

Hweemin Tan: Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Kerry Kawakami:** Writing – review & editing, Conceptualization. **Jinpeng Fu:** Writing – original draft, Formal analysis. **Shuoan Li:** Methodology, Formal analysis. **Zhihe Pan:** Methodology, Formal analysis. **Xia Fang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Funding

This work was supported by the National Natural Science Foundation of China (32300908) and the Hangzhou Natural Science Foundation (2025SZRJJ1841).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank Jiaying Lu for her valuable contributions in the data collection and coding, and Disa Sauter for her insightful comments on the manuscript. This work was supported by the Zhejiang Key Laboratory of Neurocognitive Development and Mental Health (Grant No. 2025E10037).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jesp.2026.105015.

References

Aviezer, H., Bentin, S., Dudarev, V., & Hassin, R. R. (2011). The automaticity of emotional face-context integration. *Emotion, 11*(6), 1406–1414. https://doi.org/10.1037/a0023578

Aviezer, H., Hassin, R. R., Ryan, J., Grady, C., Susskind, J., Anderson, A., … Bentin, S. (2008). Angry, disgusted, or afraid? Studies on the malleability of emotion perception. *Psychological Science, 19*(7), 724–732. https://doi.org/10.1111/j.1467-9280.2008.02148.x

Bachorowski, J.-A., & Owren, M. J. (1995). Vocal expression of emotion: Acoustic properties of speech are associated with emotional intensity and context. *Psychological Science, 6*(4), 219–224. https://doi.org/10.1111/j.1467-9280.1995.tb00596

Baltrusaitis, T., Zadeh, A., Lim, Y. C., & Morency, L.-P. (2018). *OpenFace 2.0: Facial behavior analysis toolkit.* . 2018 13th IEEE international conference on Automatic Face & Gesture Recognition (FG 2018), (pp. 59–66). https://doi.org/10.1109/FG.2018.00019.

Bao, H. (2024). *bruceR: Broadly useful convenient and efficient R functions (R package version 2024.6).* . CRAN. https://doi.org/10.32614/CRAN.package.bruceR.

Bates, D., Maechler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software, 67*(1), 1–48. https://doi.org/10.18637/jss.v067.i01

Brooks, J. A., & Freeman, J. B. (2018). Conceptual knowledge predicts the representational structure of facial emotion perception. *Nature Human Behaviour, 2*(8), 581–591. https://doi.org/10.1038/s41562-018-0376-6

Campanella, S., & Belin, P. (2007). Integrating face and voice in person perception. *Trends in Cognitive Sciences, 11*(12), 535–543. https://doi.org/10.1016/j.tics.2007.10.001

Chen, P., Chung-Fat-Yim, A., Guo, T., & Marian, V. (2024). Cultural background and input familiarity influence multisensory emotion perception. *Cultural Diversity & Ethnic Minority Psychology, 30*(3), 487–496. https://doi.org/10.1037/cdp0000577

Chen, P., Chung-Fat-Yim, A., & Marian, V. (2022). Cultural experience influences multisensory emotion perception in bilinguals. *Languages, 7*(1), 12. https://doi.org/10.3390/languages7010012

Collignon, O., Girard, S., Gosselin, F., Roy, S., Saint-Amour, D., Lassonde, M., & Lepore, F. (2008). Audio-visual integration of emotion expression. *Brain Research, 1242*, 126–135. https://doi.org/10.1016/j.brainres.2008.04.023

Cordaro, D. T., Sun, R., Keltner, D., Kamble, S., Huddar, N., & McNeil, G. (2018). Universals and cultural variations in 22 emotional expressions across five cultures. *Emotion (Washington, D.C.), 18*(1), 75–93. https://doi.org/10.1037/emo0000302

Crivelli, C., & Fridlund, A. J. (2018). Facial displays are tools for social influence. *Trends in Cognitive Sciences, 22*(5), 388–399. https://doi.org/10.1016/j.tics.2018.02.006

Ekman, P., Friesen, W. V., & Hager, J. V. (2002). *Facial action coding system* (2nd ed.). Research Nexus Ebook.

Elfenbein, H. A., & Ambady, N. (2002). On the universality and cultural specificity of emotion recognition: A meta-analysis. *Psychological Bulletin, 128*(2), 203–235. https://doi.org/10.1037/0033-2909.128.2.203

Eyben, F., Wöllmer, M., & Schuller, B. (2010). *OpenSMILE: The Munich versatile and fast open-source audio feature extractor*. . Proceedings of the 18th ACM international conference on multimedia, (pp. 1459–1462). https://doi.org/10.1145/1873951.1874246.

Fang, X., & Kawakami, K. (2024). Decoding angry and disgusted faces across cultures: Facial prototypes and software matter. *Journal of Nonverbal Behavior, 48*(3), 439–464. https://doi.org/10.1007/s10919-024-00453-0

Fang, X., Sauter, D. A., Heerdink, M. W., & Van Kleef, G. A. (2022). Culture shapes the distinctiveness of posed and spontaneous facial expressions of anger and disgust. *Journal of Cross-Cultural Psychology, 53*(5), 471–487. https://doi.org/10.1177/00220221221095208

Fang, X., Sauter, D. A., & Van Kleef, G. A. (2018). Seeing mixed emotions: The specificity of emotion perception from static and dynamic facial expressions across cultures. *Journal of Cross-Cultural Psychology, 49*(1), 130–148. https://doi.org/10.1177/0022022117736270

Fang, X., van Kleef, G. A., & Sauter, D. A. (2019). Revisiting cultural differences in emotion perception between easterners and westerners: Chinese perceivers are accurate, but see additional non-intended emotions in negative facial expressions. *Journal of Experimental Social Psychology, 82*, 152–159. https://doi.org/10.1016/j.jesp.2019.02.003

Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*(2), 175–191. https://doi.org/10.3758/bf03193146

Gao, C., Oh, S., Yang, X., Stanley, J. M., & Shinkareva, S. V. (2024). Neural representations of emotions in visual, auditory, and modality-independent regions reflect idiosyncratic conceptual knowledge. *Human Brain Mapping, 45*(14), Article e70040. https://doi.org/10.1002/hbm.70040

de Gelder, B., & Vroomen, J. (2000). The perception of emotion by ear and by eye. *Cognition and Emotion, 14*(3), 289–311. https://doi.org/10.1080/026999300378824

de Gelder, B., Vroomen, J., Pourtois, G., & Weiskrantz, L. (2000). Affective blindsight: Are we blindly led by emotions? Response to Heywood and Kentridge (2000). *Trends in Cognitive Sciences, 4*(4), 126–127. https://doi.org/10.1016/s1364-6613(00)01470-4

Gil, S., Hattouti, J., & Laval, V. (2016). How children use emotional prosody: Crossmodal emotional integration? *Developmental Psychology, 52*(7), 1064–1072. https://doi.org/10.1037/dev0000121

Goudbeek, M., & Scherer, K. (2010). Beyond arousal: Valence and potency/control cues in the vocal expression of emotion. *The Journal of the Acoustical Society of America, 128*(3), 1322–1336. https://doi.org/10.1121/1.3466853

Holler, J., & Levinson, S. C. (2019). Multimodal language processing in human communication. *Trends in Cognitive Sciences, 23*(8), 639–652. https://doi.org/10.1016/j.tics.2019.05.006

Jack, R. E., Blais, C., Scheepers, C., Schyns, P. G., & Caldara, R. (2009). Cultural confusions show that facial expressions are not universal. *Current Biology, 19*(18), 1543–1548. https://doi.org/10.1016/j.cub.2009.07.051

Jack, R. E., Sun, W., Delis, I., Garrod, O. G., & Schyns, P. G. (2016). Four not six: Revealing culturally common facial expressions of emotion. *Journal of Experimental Psychology: General, 145*(6), 708–730. https://doi.org/10.1037/xge0000162

Kassambara, A. (2023). *Rstatix: Pipe-friendly framework for basic statistical tests (R package version 0.7.2)*. . CRAN. https://doi.org/10.32614/CRAN.package.rstatix.

Kassambara, A. (2025). *Ggpubr: 'ggplot2' based publication ready plots (R package version 0.6.1)*. . CRAN. https://doi.org/10.32614/CRAN.package.ggpubr.

Kawahara, M., Sauter, D. A., & Tanaka, A. (2021). Culture shapes emotion perception from faces and voices: Changes over development. *Cognition and Emotion, 35*(6), 1175–1186. https://doi.org/10.1080/02699931.2021.1922361

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software, 82*(13), 1–26. https://doi.org/10.18637/jss.v082.i13

Laukka, P., & Elfenbein, H. A. (2021). Cross-cultural emotion recognition and in-group advantage in vocal expression: A meta-analysis. *Emotion Review, 13*(1), 3–11. https://doi.org/10.1177/1754073919897295

Laukka, P., Elfenbein, H. A., Thingujam, N. S., Rockstuhl, T., Iraki, F. K., Chui, W., & Althoff, J. (2016). The expression and recognition of emotions in the voice across five nations: A lens model analysis based on acoustic features. *Journal of Personality and Social Psychology, 111*(5), 686–705. https://doi.org/10.1037/pspi0000066

Laukka, P., Juslin, P., & Bresin, R. (2005). A dimensional approach to vocal expression of emotion. *Cognition and Emotion, 19*(5), 633–653. https://doi.org/10.1080/02699930441000445

Lecker, M., Dotsch, R., Bijlstra, G., & Aviezer, H. (2020). Bidirectional contextual influence between faces and bodies in emotion perception. *Emotion, 20*(7), 1154–1164. https://doi.org/10.1037/emo0000619

Lee, L. H., Li, J. H., & Yu, L. C. (2022). Chinese EmoBank: Building valence-arousal resources for dimensional sentiment analysis. *Transactions on Asian and Low-Resource Language Information Processing, 21*(4), 1–18. https://doi.org/10.1145/3489141

Lenth, R. V. (2025). *Emmeans: Estimated marginal means, aka least-squares means (R package version 1.11.2)*. . CRAN. https://doi.org/10.32614/CRAN.package.emmeans.

Liu, P., Rigoulot, S., Jiang, X., Zhang, S., & Pell, M. D. (2021). Unattended emotional prosody affects visual processing of facial expressions in mandarin-speaking Chinese: A comparison with English-speaking Canadians. *Journal of Cross-Cultural Psychology, 52*(3), 275–294. https://doi.org/10.1177/0022022121990897

Liu, P., Rigoulot, S., & Pell, M. D. (2015a). Cultural differences in on-line sensitivity to emotional voices: Comparing east and west. *Frontiers in Human Neuroscience, 9*, Article 311. https://doi.org/10.3389/fnhum.2015.00311

Liu, P., Rigoulot, S., & Pell, M. D. (2015b). Culture modulates the brain response to human expressions of emotion: Electrophysiological evidence. *Neuropsychologia, 67*, 1–13. https://doi.org/10.1016/j.neuropsychologia.2014.11.034

Massaro, D. W., & Egan, P. B. (1996). Perceiving affect from the voice and the face. *Psychonomic Bulletin & Review, 3*(2), 215–221. https://doi.org/10.3758/BF03212421

McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. *Nature, 264*(5588), 746–748. https://doi.org/10.1038/264746a0

Niedenthal, P. M., & Brauer, M. (2012). Social functionality of human emotion. *Annual Review of Psychology, 63*, 259–285. https://doi.org/10.1146/annurev.psych.121208.131605

Pan, Z., Tan, H., Liu, S., & Fang, X. (2026). Beyond the basic six, static, and WERID: Exploring the range of emotions conveyed by facial expressions. *Journal of Experimental Social Psychology, 122*, Article 104836. https://doi.org/10.1016/j.jesp.2025.104836

Peirce, J. W., Gray, J. R., Simpson, S., MacAskill, M. R., Höchenberger, R., Sogo, H., … Lindeløv, J. (2019). Psychopy2: Experiments in behavior made easy. *Behavior Research Methods*. https://doi.org/10.3758/s13428-018-01193-y

Pell, M. D., Paulmann, S., Dara, C., Alasseri, A., & Kotz, S. A. (2009). Factors in the recognition of vocally expressed emotions: A comparison of four languages. *Journal of Phonetics, 37*(4), 417–435. https://doi.org/10.1016/j.wocn.2009.07.005

Puffet, A. S., Saleh, G., & Rigoulot, S. (2025). Exploring the impact of facial-body emotion incongruence: Insight from a systematic review. *Journal of Nonverbal Behavior, 49*(3), 345–376.

R Core Team. (2024). *R: A language and environment for statistical computing [computer software, version 4.3.3]*. Vienna, Austria: R Foundation for Statistical Computing.

Sauter, D. A., Eisner, F., Ekman, P., & Scott, S. K. (2010). Cross-cultural recognition of basic emotions through nonverbal vocalizations. *Proceedings of the National Academy of Sciences, 107*(6), 2408–2412. https://doi.org/10.1073/pnas.0908239106

Scarantino, A. (2017). How to do things with emotional expressions: The theory of affective pragmatics. *Psychological Inquiry, 28*(2–3), 165–185. https://doi.org/10.1080/1047840X.2017.1328951

Scherer, K. R. (2018). Acoustic patterning of emotion vocalizations. In S. Frühholz, & P. Belin (Eds.), *The Oxford handbook of voice perception* (pp. 61–92). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198743187.013.4.

Scherer, K. R., Banse, R., & Wallbott, H. G. (2001). Emotion inferences from vocal expression correlate across languages and cultures. *Journal of Cross-Cultural Psychology, 32*(1), 76–92. https://doi.org/10.1177/0022022101032001009

Susskind, J. M., Littlewort, G., Bartlett, M. S., Movellan, J., & Anderson, A. K. (2007). Human and computer recognition of facial expressions of emotion. *Neuropsychologia, 45*(1), 152–162. https://doi.org/10.1016/j.neuropsychologia.2006.05.001

Takagi, S., Hiramatsu, S., Tabei, K., & Tanaka, A. (2015). Multisensory perception of the six basic emotions is modulated by attentional instruction and unattended modality. *Frontiers in Integrative Neuroscience, 9*, Article 1. https://doi.org/10.3389/fnint.2015.00001

Tanaka, A., Koizumi, A., Imai, H., Hiramatsu, S., Hiramoto, E., & de Gelder, B. (2010). I feel your voice. Cultural differences in the multisensory perception of emotion. *Psychological Science, 21*(9), 1259–1262. https://doi.org/10.1177/0956797610380698

Trujillo, J. P., & Holler, J. (2023). Interactionally embedded gestalt principles of multimodal human communication. *Perspectives on Psychological Science, 18*(5), 1136–1159. https://doi.org/10.1177/17456916221141422

Van Kleef, G. A. (2009). How emotions regulate social life: The emotions as social information (EASI) model. *Current Directions in Psychological Science, 18*(3), 184–188. https://doi.org/10.1111/j.1467-8721.2009.01633.x

Van Kleef, G. A. (2016). *The interpersonal dynamics of emotion: Toward an integrative theory of emotions as social information*. Cambridge University Press. https://doi.org/10.1017/CBO9781107261396

Van Kleef, G. A., & Côté, S. (2022). The social effects of emotions. *Annual Review of Psychology, 73*, 629–658. https://doi.org/10.1146/annurev-psych-020821-010855

Vroomen, J., Driver, J., & de Gelder, B. (2001). Is cross-modal integration of emotional expressions independent of attentional resources? *Cognitive, Affective, & Behavioral Neuroscience, 1*(4), 382–387. https://doi.org/10.3758/cabn.1.4.382

Watson, R., Latinus, M., Noguchi, T., Garrod, O., Crabbe, F., & Belin, P. (2014). Crossmodal adaptation in right posterior superior temporal sulcus during face–voice emotional integration. *The Journal of Neuroscience, 34*(20), 6813–6821. https://doi.org/10.1523/JNEUROSCI.4478-13.2014

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis (R package version 3.5.2)*. Springer-Verlag. https://ggplot2.tidyverse.org.